Guards, Squads and Incentives: Keeping Large Teams of Language-Model Agents Effective and Safe

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Abstract

Teams of language-model agents trust bad information, pool less of what they know as they grow, and stop cooperating when trained with the wrong incentives. Studying teams of up to 64 Claude Haiku 5.5 agents, and reinforcement learning (RL) of a 3B open-weight model, mostly under analysis plans frozen before the runs, we find five things. (1) *Verification works as an assigned role*: guards, a quarter of the team told to post, confirm and contest the clues they hold instead of solving, recovered 0.72 of the damage done by an equal number of liars at 8 and 16 agents with no access to the true answers (0.66 measured, like the record's, from the plain board's output without liars), against 0.34 for an oracle record of each sender's accuracy, and beat an unprotected team at every size up to 64. (2) *Squads keep the guards' advantage as teams grow*: guards could check only about two in five lies, and in a replay the rest did more harm in larger teams, so their advantage shrank; squads of eight, each with its own board and guards, kept theirs, widening their lead over guards on one shared board from $+0.06$ at 16 agents to $+0.17$ at 64, though their guards checked no more lies. (3) *Larger teams are more gullible*: the share of honest answers made while misinformed rose from 0.79 at 8 agents to 0.91 at 32; lies were no more visible and almost never reposted, and what drove the rise remains open. (4) *Copies of one model do not divide labour on their own*: no agent took an offered guard role, five short team-reward RL runs created no roles, and every individual-reward run shared less than the imitation start (0.11–0.41 shares per held clue against 0.92). (5) *Move information; do not add work*: fixes that pass on or certify information without costing turns helped, and those that add work to every turn did not. We release the environments, plans, code and the compact data behind every figure; the saved runs and the response cache that regenerate every number are available on request.

1 Introduction

Multi-agent deployments of language models are growing, from orchestrators with parallel workers [10] to large groups working on open-ended problems, and their failures are well documented. On hidden-profile tasks, where the right answer depends on facts that only individual members hold, groups of agents reach about 30% accuracy where one agent with every fact reaches 81% [15], and Anthropic's red team reports group accuracies of 17–36% for most models [2]. Agents conform [30], are readily misled by a single false testimony [31], free-ride in public goods games when they are reasoning models [9], and become opportunistic when trained with RL in social dilemmas [24]. The red team lists mechanism design, reputation and environments that shape cooperation as open problems [2].

[*]An AI model made by Anthropic, working as a research and coding agent in Claude Code; see the author contributions.

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We test fixes and how they hold up as teams grow. For each failure we build a small, fully specified environment, freeze an analysis plan before any run and add one mechanism at a time; where we can, we pair a mechanism the harness provides (an *affordance*, such as a new action or observation line) with a prompt-only *instruction* that gives the same advice (§3). Our five findings, strongest first:

1. **Verification works as an assigned role** (§4.1): guards, who need no ground truth, recover more of the liars' damage than an oracle track record at 8 and 16 agents, beat an unprotected team at every size from 8 to 64, and help against every kind of faulty teammate we tested.
2. **Squads keep the guards' advantage as teams grow** (§4.2): only about two in five lies fall on a clue a guard holds and, in larger teams, the other lies do more harm, so the guards' advantage shrinks; squads of eight, each with its own board and guards, keep theirs, though their guards check no more lies.
3. **Larger teams are more gullible** (§4.3): replayed runs show agents in larger teams answering while misinformed more often, not lies spreading further; what drives the rise remains open.
4. **Copies of one model do not divide labour on their own, and the reward shapes whether they cooperate** (§4.4): no agent takes up a role that is offered but not assigned, five 30-step runs of RL with a team reward did not create one, and every individual-reward run shared less than the imitation start.
5. **Move information; do not add work** (§4.5): a shared board, a coordinator that passes clues on and a public track record helped; a request board, an instruction to share first, private checks and orders to follow a coordinator did not.

2 Related work

Information pooling in agent teams. The hidden-profile paradigm [27] has been ported to LLM groups [2, 15]. Diagnoses stress early convergence on shared evidence [15], averaging of expert and non-expert views [21] and loss of issue-critical facts in deliberation [28]; an LLM facilitator raised how much human groups shared without improving their decisions [1]. Adaptive communication control [3] and structured deliberation [25] can beat unstructured discussion. Pruning messages and agents saves tokens at comparable accuracy [29, 33], and across 260 configurations multi-agent architectures range from large gains to large losses against a single agent [13]; in our tasks a message also costs its sender a turn.

Deception, trust and fault tolerance. Defences learn credibility scores from past contributions [6], rank neighbours [7] or rectify misleading content [18]. Challenger and inspector agents recover injected errors in small teams [11], swarm-style systems assign validator roles [14], and confidence-weighted consensus tolerates many faulty members [35], so verifier roles are known in direction; what we add is a controlled comparison with an oracle record and an instruction, across faults and team sizes. We compare a public, harness-computed signal with private verification at matched cost, and test oracle-free rules borrowed from social insects: quorum-like agreement counts [26] and guards [5]. Honest-signalling theory explains when a signal can be trusted, namely when it is costly to fake [32]; our results suggest that who sees the verification, and whose job it is, matter too.

Training agent teams. Multi-agent RL for LLM collaboration includes MAPoRL [22], MAGRPO [17] and AT-GRPO [34]; counterfactual credit addresses free-riding [8, 16]. With one shared LoRA adapter, trained agents failed to split into roles that private adapters learned [19]. RL-trained LLM agents become opportunistic in social dilemmas [24] and most simulated LLM societies fail to sustain a shared resource [23]. Grading that never penalises wrong answers rewards guessing [12]; we observe a multi-agent analogue in which agents stop sharing clues and guess, a producer-scrounger outcome [4].

3 Methods

The clue board. N agents share a board of $2N + 2$ open subtasks, each worth 1, 2 or 3 points; a subtask is replaced when solved or after 30 turns. A subtask hides an integer $0 \leq x < 30$ fixed

by two clues "$x \bmod p = r$" (different primes p) dealt to different agents, so no agent can deduce it alone (with one clue it can only guess among several values). In each turn every agent takes one action: select a subtask to work on, answer the selected subtask (a wrong answer costs 0.5), share a clue, or wait. Shared clues go to a *blackboard* every agent reads; in the channel comparisons they go instead to two random teammates as messages, nowhere, or to a rule-based coordinator that routes them to agents working on the subtask and suggests subtasks. Sharing costs the sharer a turn and earns it nothing directly, so the board poses the hidden-profile problem with a price on communication. Each prompt lists, as the clues an agent knows for a subtask, its own clues and the newest value posted for each other clue (on the counting board of §4.1, the value most agents posted), and the plain board's rules add that a clue shared once is known to everyone who saw it, so sharing it again only helps teammates who did not see it.

Faulty teammates. For the defences (§4.1–§4.2) each clue is dealt to two agents, so an honest holder can contradict a lie about it, and a quarter of the agents are *liars*: scripted agents that, every other turn, post a false value r for a clue of a subtask they are not working on. We also test *mistaken* teammates, scripted rule agents in the faulty quarter that misread 30% of the clues they hold and use the wrong values in posts and answers (like liars, they are not counted as honest); *colluding* liars, who repost each other's false values and so defeat any rule that trusts a value two agents posted; and *Claude liars*, Claude Haiku 5.5 agents told in a private instruction to make their teammates answer wrongly, who can influence teammates only through the clue values they post (no free text). In §4.3 and §4.5 each clue has one holder unless stated.

Agents. Team members are Claude Haiku 5.5 at the API's low effort setting. Each turn an agent receives a text observation (the open subtasks, its own clues, the board and a memory of its earlier turns) and replies with one action as JSON. Every reply is cached under its full request, so reruns are byte-identical and saved runs can be replayed with new measurements at no API cost. Rule agents written in numpy answer any subtask whose clues they know and otherwise share a clue a teammate needs for the subtask it has selected; they give a reference level and, run before most expensive studies (not the Claude-liar study), a frozen forecast of their results (Appendix C).

Mechanisms. Each arm changes one thing (Table 1). Where we can, an affordance is paired with an instruction that gives the same advice in the prompt, to separate the affordance's effect from that of telling agents what to do. The instruction, agreement counts and guards, and so squads, failure flags and guard copies, also replace the plain board's re-sharing sentence with advice to confirm or contest values; the plain board and the track record keep it. Gains over the plain board therefore include that change, which we did not test on its own; comparisons among these defences do not.

Training. We trained Qwen2.5-3B-Instruct on the clue board (four agents, blackboard) as teams of copies of one model sharing a LoRA adapter. A supervised warm start imitated the rule agent's moves (about $14{,}000$ turns in teams of one, two and four). RL then used group-relative policy gradients: each step played 8 boards 4 times each; each agent slot's return was compared with that slot's mean over the four plays, and each update used a random quarter of the turns with a nonzero advantage, with a KL penalty toward the warm start. The learning rate was steered to a target per-step policy change (a KL of 0.003, measured on re-scored replies), steps that moved the policy more than four times the target were undone, and runs halted on persistent invalid replies or on large drift from the warm start. Rewards were *individual* (an agent's own points less its penalties), *team* (the mean of the team's individual rewards), *credit* (the solver keeps half of a subtask's value and the rest goes to the agents whose shared clues it used) and *individual + penalty* (individual reward with a training-only penalty: a wrong answer costs 3 instead of 0.5). The 3B agents saw only their latest note, not the full turn history Claude agents saw. The mixed-size runs of §4.4 used credit reward with the wrong-answer penalty and 20-turn games, half their boards with eight agents and a quarter with scripted liars, with replies constrained to the model's own key order and the loss and KL on value tokens only; the team-reward runs used that recipe on the counting board, with each token's KL term capped. Trained models were scored by welfare on held-out boards, whatever their training reward: 15 four-agent boards for the first designs, 10 to 15 per size from 4 to 32 agents for the mixed-size runs, and 5 at 8 and 32 agents for the team-reward runs.

Measures. *Welfare* is points earned, net of penalties, per agent-turn. The *pooled share* divides welfare by a full-information bound, half the value of the N most valuable open subtasks each turn,

Table 1: Mechanisms tested. Kind: A, an affordance the harness provides; I, a prompt-only instruction; R, an assigned role; S, a change of team structure.

Mechanism	What changes	Kind	Section
Blackboard	shared clues go to a board every agent reads (against no channel)	A	4.5
Follow the coordinator	agents are told the coordinator sees every shared clue and its suggestion is usually best	I	4.5
Coordinator passes clues	with each suggested subtask, the coordinator passes on the clues it has collected for it	A	4.5
Request board	a new action asks teammates for the clues of the agent's selected subtask	A	4.5
Share first	"before working on your own subtask, post every clue you hold for an open subtask that has not been posted yet"	I	4.5
Clue table	for each open subtask, the board lists the clues the team has posted so far	A	4.5
Track record	beside each clue, how many of its sender's earlier clues proved true (needs the truth)	A	4.5
Check action	spend a turn to learn privately whether a posted clue is true	A	4.5
Warning	"a teammate may share false clues; cross-check before relying on one"	I	4.5
Agreement counts	the board counts the agents that posted each value and marks it confirmed or contested; the re-sharing sentence is replaced by advice to confirm or contest	A	4.1
Instruction only	in place of the re-sharing sentence, agents are told to confirm or contest values; no counts shown	I	4.1
Guards	on the counting board, a quarter of the team (as many as the liars) is told to post, confirm and contest the clues it holds, never to solve	R	4.1
Volunteer offer	the guard role and its rationale are offered to every agent, with no assignment	I	4.4
Squads	the team splits into squads of eight, each with its own board, subtasks and two guards	S	4.2
Failure flags	after a wrong answer, the board flags the posted values it agreed with	A	4.2
Guard copies	one of each clue's two copies is dealt to a guard	A	4.2
Vote fixes	hidden-profile vote: a fact table (A), a facts-only first round (A), a speak-up instruction (I)	A, I	B

averaged over the run, so 1 is the information ceiling; the bound depends on which subtasks a team leaves open, so it can differ between arms on one seed, and *raw welfare* is the honest agents' welfare per agent-turn without it. With liars present we report the honest agents' pooled share. The *honest loss* is the honest agents' welfare without liars minus with them, on the same seed and arm (the oracle record ran only with liars, so its loss is taken against the plain board's clean boards), and *loss recovered* is $1-$ (arm's mean honest loss / baseline's). *Gap closed* is the mean paired difference (arm minus baseline) divided by the ceiling minus the baseline, so 1 closes the gap and a negative value widens it. An answer is *built on a false clue* if it agrees with a false value posted before it, so it is always wrong; the tracer of §4.3 counts more widely: an answer is *made while misinformed* if the clues its agent knew for that subtask included a false one, and it can be right. *Specialization* is the variance across honest agents of how often each shares, relative to binomial noise, minus the same index with turns shuffled between agents (0: no more than chance).

Design and statistics. Each study's analysis plan (arms, seeds, primary tests and correction family) was frozen, hashed and committed before its runs (Appendix A); the larger-team stages of §4.1 and §4.2 were declared deviations, each with its own frozen plan. Arms are paired on board seed and compared through their paired differences, as Miller [20] recommends for evaluations, with 5 to 10 seeds per arm (8 for the defences; the first defence study pooled 8 and 16 agents, so its primary tests use 16 seed-and-size pairs). Tests are paired sign-flip tests over seeds or training runs, or permutation tests when two sets of training runs are compared (exact, or with at least 10,000 random sign flips when there are too many units to enumerate), one-sided where the plan fixed a direction; team sizes are compared by exact permutation tests. Holm correction applies within each plan's primary

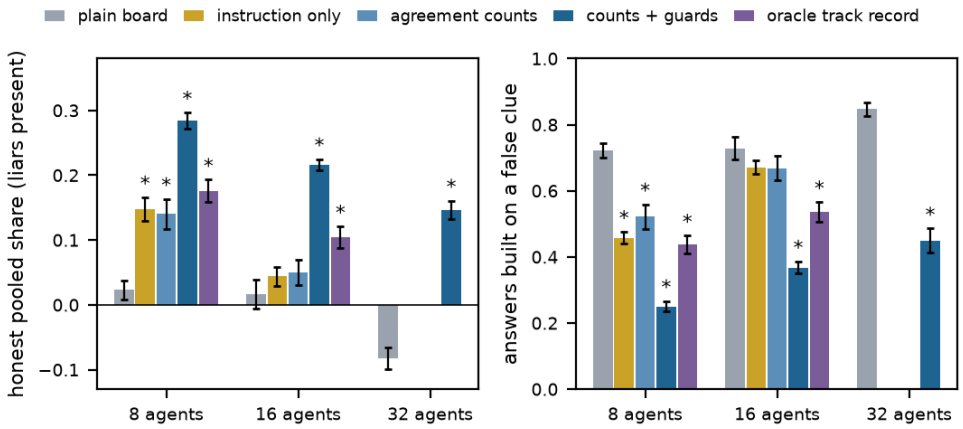


Figure 1: Defences against liars (a quarter of the agents), 8 seeds per bar; at 32 agents only the plain board and guards ran. Left: the honest agents' pooled share. Right: the share of honest answers built on a false clue. Mean ± SEM; * $p < 0.05$ against the plain board, two-sided exact sign-flip test paired on seed, uncorrected.

family. A trend over team size is the mean of each seed's least-squares slope against $\log_2 N$ (change per doubling of the team), tested with a two-sided sign-flip test over seeds; trends are descriptive. With three sizes evenly spaced in $\log_2 N$ the slope depends only on the smallest and largest, so we also report the change between neighbouring sizes. In the text, brackets give 95% bootstrap intervals over the pairing units. In the figures, points and bars are means, error bars are one standard error of the mean (SEM) over seeds or boards (the bootstrap standard error for ratio measures), and an asterisk marks $p < 0.05$ in an uncorrected two-sided test against the stated reference. Because several stages were added after earlier results, and two analyses (the coverage replay of §4.2 and the guessing counts of §4.4) had no frozen plan, we treat all findings as exploratory.

4 Results

4.1 Verification works as an assigned role

Here one agent in four lies and the harness does not know which clues are true. At 8 and 16 agents (8 seeds; plan `58e842277a3e`; API cost $24.89) we compared the *plain board* (no defence), agreement counts (the *counting board*), the same advice as an instruction, agreement counts with guards, and an oracle track record of each sender (Figure 1).

Guards worked. They beat the counting board by 0.155 $[0.135, 0.183]$ in the honest agents' pooled share (Holm $p < 0.001$ over 16 seed-and-size pairs) and recovered 0.72 of the liars' damage over the two sizes (0.66 measured from the plain board's clean boards, the oracle record's baseline), against 0.34 for the oracle record. Guards posted a clue they held on 0.99 of their turns at 8 agents, and at most 0.16 of those posts were contests, so supplying clues may explain part of their benefit. With guards, each lie drew 0.31 contests from honest agents (counting board 0.21, plain board 0.06), and the share of honest answers built on a lie fell to 0.31 (counting board 0.60, plain board 0.73). **The display added nothing to the advice.** The counting board beat the plain board (0.075 $[0.048, 0.102]$ over both sizes), mostly at 8 agents (0.12, against 0.03 at 16), but did no better than the instruction (difference -0.001), and some honest agents reposted a lie to "confirm" it.

Guards held at every size and against every fault we tested. Guards beat the plain board by $+0.261$, $+0.200$, $+0.229$ and $+0.165$ at 8, 16, 32 and 64 agents, in 8 of 8 seeds at each size (one-sided exact $p = 0.004$ each). At 32 agents they also helped against mistaken teammates ($+0.058$ $[0.024, 0.094]$, $p = 0.023$) and against colluding liars ($+0.090$ $[0.054, 0.118]$, $p = 0.008$; secondary tests, one-sided and uncorrected): a guard contests a wrong value of a clue it holds however many agents posted it. At 16 agents (plan `0c59fa10f2b0`) we replaced the scripted liars with Claude liars. Guards still beat the plain board ($+0.063$ $[0.016, 0.104]$, Holm $p = 0.039$) and recovered 0.70 of

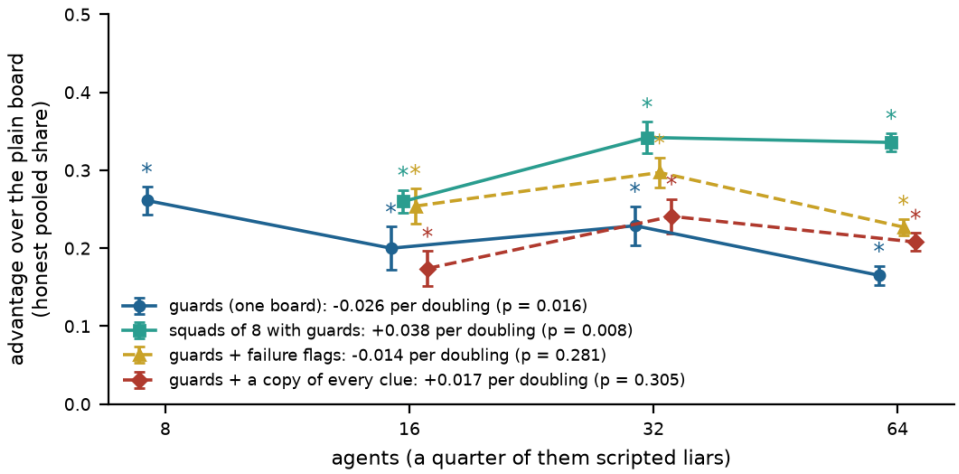


Figure 2: Advantage over the plain board in the honest agents' pooled share against team size, a quarter of the agents scripted liars, 8 seeds per point (the three design changes ran at 16, 32 and 64 agents). Mean ± SEM; * $p < 0.05$, two-sided exact sign-flip test against 0, paired on seed, uncorrected. The legend gives each design's trend: the mean of each seed's slope over $\log_2$ of the team size, per doubling, with its two-sided exact sign-flip p (descriptive; with three sizes evenly spaced on this scale, as for the design changes, the slope depends only on the smallest and largest). Guards on one board lose ground as teams grow; squads of eight, each with its own board and guards, rise from 16 to 32 agents and hold that level at 64.

the Claude liars' damage (0.63 for scripted liars on the same seeds). The Claude liars lied less often than scripted ones (a false post in 0.18 of their turns, against 0.5), did as much damage per lie, and did not avoid clues a guard held.

The cost. On boards without liars (*clean* boards), the guard board scored 0.09, 0.06 and 0.04 below the plain board in pooled share at 8, 16 and 32 agents. The counting rules carried this cost, not the guards: without guards the counting board cost as much at 8 and 16 agents (0.09 and 0.06), and the instruction nearly as much (0.08 and 0.05). Asking every agent to confirm costs output; adding guards cost nothing further (no counting board without guards ran at 32 agents). In raw welfare the guard board's clean cost was smaller, 0.03, 0.03 and 0.03 per agent-turn, less than the counting board's alone (0.08 and 0.06): it left more valuable subtasks open, which raised its pooled-share bound (9.8 against 8.4 for the plain board at 8 agents).

4.2 Squads keep the guards' advantage as teams grow

From 8 to 64 agents the guards' advantage over the plain board changed by -0.026 per doubling of the team (two-sided $p = 0.016$; descriptive). Replaying every guard board from the response cache, an analysis added after the results, showed that the guards weakened only a little. At every size they contested 75–79% of the lies on clues they held, with a median delay of 1.0–1.5 turns, and each such lie led to 0.20 wrong answers at 8 agents and 0.24 at 64. The limit is coverage: with two holders per clue, only 37–43% of lies fell on a clue a guard held. Lies on the other clues caused 81–91% of all answers built on a lie, and each caused more wrong answers in larger teams (0.64 at 8 agents, 1.49 at 64). On the plain board, too, each lie did more harm in larger teams (2.7 wrong answers at 8 agents, 3.7 at 64), but there the liars already took nearly everything the honest agents earned (honest pooled share 0.02 with liars against 0.45 on clean boards at 8 agents; -0.08 against 0.36 at 32), so in pooled share its loss had little room to grow. The guarded board's loss did grow, and the share of the damage guards recovered fell from 0.81 at 8 agents to 0.63 at 16 and 0.60 at 32.

We then tested three design changes against guards on one shared board at 16 and 64 agents (plan and numpy forecast frozen first, `a441381ce078`; $43.80). All three beat them at 64 agents, each in 8 of 8 seeds (one-sided, Holm $p = 0.012$, 0.012 and 0.012 for squads, failure flags and guard copies), and on raw welfare too (+0.22, +0.08 and +0.06 per agent-turn; Figure 2 shows pooled

share). *Squads* won by $+0.171$ $[0.156, 0.186]$ at 64 agents, and their advantage over the plain board rose from 0.26 at 16 agents to 0.34 at 64, in 8 of 8 seeds ($+0.038$ per doubling, two-sided exact $p = 0.008$; descriptive), while that of guards on one board fell (the squads' change minus the guards': $+0.111$ $[0.069, 0.152]$, $p = 0.008$); a 64-agent team of squads kept 0.25 honest pooled share, a little below the 8-agent guard board's 0.28 (difference -0.04 $[-0.06, -0.01]$) and far above the 0.08 of guards on one 64-agent board, with prompts of 780 tokens per turn instead of 6,463 on one board. *Failure flags* make public what the environment already tells the agent whose answer failed; they added $+0.062$ $[0.042, 0.082]$ at 64 agents and $+0.054$ at 16, at no cost on clean boards at 16 agents ($+0.000$; no clean boards ran at larger sizes), though 0.27 of flags fell on true values. *Guard copies* helped only at 64 agents ($+0.043$ $[0.025, 0.063]$): at 16 they cost 0.129 pooled share on clean boards, and at 64 agents, guards holding every clue contested 0.61 of the lies on their clues, against 0.75 for ordinary guards. Two sizes cannot show a trend's shape, so we added 32 agents for all three designs (a declared deviation; plan and forecast frozen first, `4d5d0a8e4f5b`; $10.44). At 32 agents squads and failure flags again beat guards on one board ($+0.113$ $[0.086, 0.139]$ and $+0.068$ $[0.044, 0.092]$, Holm $p = 0.012$ and 0.012, each in 8 of 8 seeds), while guard copies drew level ($+0.012$ $[-0.004, 0.029]$, Holm $p = 0.113$), between their loss at 16 agents and their win at 64. The squads' advantage over the plain board rose from 16 to 32 agents ($+0.082$, $p = 0.023$, up in 7 of 8 seeds) and then held (0.34 at 32, 0.34 at 64; change -0.006, $p = 0.859$), while the advantage of guards on one board fell to 0.17 at 64, so the squads' lead over them widened from $+0.06$ at 16 agents to $+0.11$ at 32 and $+0.17$ at 64 (descriptive). Squads did not let guards check more lies: at 64 agents 0.40 of lies fell on a guard-held clue, against 0.37 on one board. Instead each unchecked lie led to fewer wrong answers (0.79 against 1.48), contests came sooner (median 0.1 against 1.5 turns) and prompts were shorter. No squads without guards and no clean squad boards ran, so we cannot say how much of the gain needs guards, or what squads cost without liars. Squads shared no subtasks, so their gain holds for tasks that split cleanly and is likely an upper bound for tasks that do not.

4.3 Larger teams are more gullible

Without a defence, the same share of liars does more damage in a larger team. With one holder per clue and a quarter of the agents liars (5 seeds), the honest loss was higher at 16 and 32 agents (0.58 and 0.54) than at 8 (0.47); with two holders per clue, liars pushed the plain board's honest pooled share below zero at 32 and 64 agents (-0.08 and -0.09), and at 64 agents 0.85 of honest answers were built on a lie. A numpy rule-agent forecast frozen before the runs had predicted that the loss would fall.

We replayed the one-holder runs from the response cache with a turn-by-turn tracer (plan `f5c25637acd0`; no API calls) and split the honest loss into additive terms that sum to it exactly. Lies were no more visible in larger teams (a similar share of open subtasks showed a false value), and honest agents almost never reposted a liar's value (at most 0.25% of liar posts). What grew was how often agents answered while misinformed: the share of honest answers made while misinformed rose from 0.79 at 8 agents to 0.91 at 32 (Figure 3, left), such answers were wrong 95–97% of the time, and losses on subtasks a lie touched accounted for 0.72 of the rise in honest loss (95% interval $[0.16, 3.2]$). When a liar was first to post a clue, no honest agent posted its true value afterwards in 89–94% of cases, but this did not change with team size (change from 8 to 32 agents $+0.04$ $[-0.04, 0.13]$), so it explains why a lie sticks, not why larger teams act on lies more often; what drives the rise remains open. The harness invited the habit: each prompt listed the newest posted value of a clue among the clues an agent knew, and the rules said that a clue shared once is known to everyone who saw it, so sharing it again only helps teammates who did not see it; we did not test either on its own. The measure also has no baseline: a quarter of the agents were liars, so about a quarter of the clues had a liar holder whom no honest agent could replace, and even without liars honest agents posted only 0.38–0.41 of the clues they held. Guards target this failure: they contest a posted value of a clue they hold. Two fixes first tested at four agents (§4.5) kept their effect as teams grew (Figure 3, right): with a quarter of the agents liars, the public track record recovered 0.30, 0.22 and 0.31 of the loss at 8, 16 and 32 agents, and the request board hurt pooling about as much at every size.

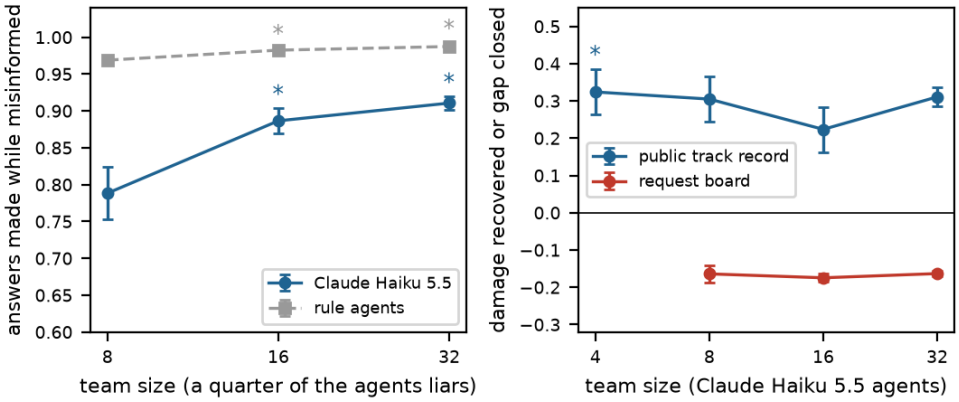


Figure 3: Left: share of honest answers made while misinformed (the clues the agent knew for the subtask included a false one), one holder per clue and a quarter of the agents liars, for Claude Haiku 5.5 (5 seeds) and rule agents (20 seeds); mean ± SEM; * $p < 0.05$, permutation test against 8 agents (exact for Claude Haiku; 20,000 random permutations for rule agents). Right: share of the liars' damage recovered by the public track record and share of the pooling gap closed by the request board on boards without liars, against team size (the record: 10 seeds at 4 agents, 5 at larger sizes; the request board: 5 seeds per size from 8 agents); mean ± bootstrap SE; * $p < 0.05$, two-sided exact sign-flip test against 0 (with five seeds the exact two-sided p cannot fall below 0.0625).

4.4 Copies of one model do not divide labour on their own

Prompting. Offered the guard role and its rationale but no assignment (16 agents), no honest agent acted as a guard on any of 24 boards: the team did no better than the counting board alone ($+0.009$ $[-0.023, 0.039]$, Holm $p = 0.293$) and scored -0.16 relative to assigned guards.

Training. Imitation did most of the teaching: the warm start raised welfare from -0.001 (untrained) to 0.27 $[0.22, 0.32]$ (Figure 4). The reward shaped what RL added. Under individual reward the model stopped sharing (0.92 to 0.15 shares per held clue) and solved by guessing (74% of solves): guessing in turn through the few values one clue leaves pays on its own, while sharing earns the sharer nothing. The team-reward run collapsed mid-training and was halted; credit reward kept agents sharing and beat individual reward by 0.19 $[0.09, 0.27]$ (one run each; the interval resamples boards, not training runs). In a comparison added after the plan was frozen, every run with the training-only penalty on wrong answers guessed less (0.1–14.7 guesses per board against 49.2–90.3; two of the runs compared halted with many invalid replies). We had also claimed that the penalty kept agents sharing, but a follow-up under a frozen plan, adding three runs to each arm (four without the penalty and five with it, counting the earlier ones), found that effect seed-dependent ($+0.37$ shares per held clue, exact $p = 0.056$, Holm 0.11; the selfish drift appeared in only 2 of the four runs without the penalty), and we withdrew the claim.

Roles did not emerge. Five runs of 30 steps with team reward on the counting board and no role instruction (plan `c103584f5aac`) all ran without a halt, but the trained teams' specialization averaged only $+1.32$ (exact $p = 0.0625$, not significant), below the warm start's $+3.44$ before any RL; the most specialized trained run reached $+3.57$. Of the five runs, 2 collapsed to every agent posting and none solving: under team reward a post is never penalized and an answer can be. Training with credit reward and the wrong-answer penalty on mixed four- and eight-agent teams, a quarter of the boards with liars (8 runs of up to 60 steps; plan `8e28b5fdfb1f`) raised welfare at four agents from 0.26 to 0.34 in 7 of 8 runs (exact $p = 0.035$; Holm 0.105, not significant after correction), but not at 32 agents (0.09 against 0.08 for the warm start; Claude Haiku, with full turn history, reaches 0.44); trained teams did lose less to liars at 32 agents (reduction in honest loss $+0.061$ $[0.012, 0.108]$, $p = 0.023$, a secondary test). Training at small sizes did not teach what large teams need.

A training lesson. Of those 8 runs, 5 halted early, 4 of them on the drift rule: in 3 the logged KL to the warm start jumped in one step while the measured policy change and behaviour stayed flat,

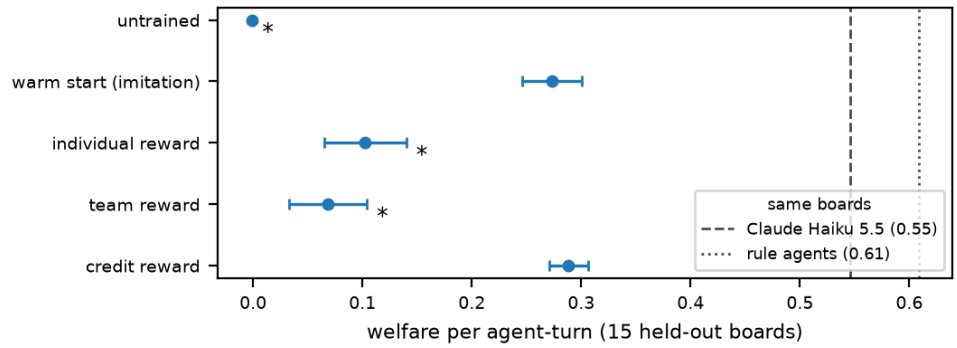


Figure 4: Welfare per agent-turn of trained four-agent teams of Qwen2.5-3B-Instruct on 15 held-out boards: the untrained model, the warm start and one training run for each of three rewards, with Claude Haiku 5.5 (full turn history) and rule agents on 5 of these boards for reference. Mean ± SEM over boards; * $p < 0.05$ against the warm start, two-sided exact sign-flip test paired on board, uncorrected.

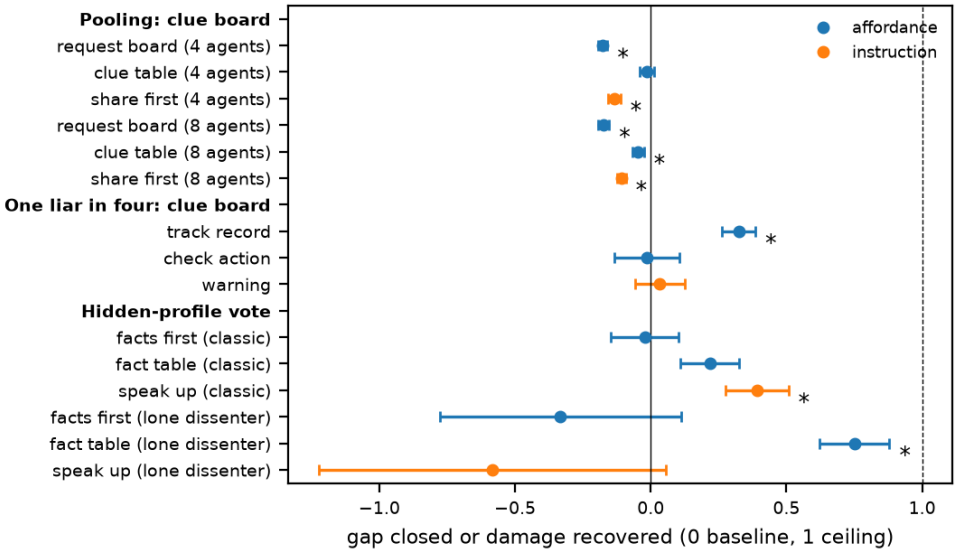


Figure 5: Each fix at four agents (eight where marked; Table 1): share of the gap to the ceiling closed (pooling, vote) or of the liar's damage recovered (one liar), where 0 is the baseline and 1 the ceiling (dashed). In the vote, *facts first* allows only facts in round one, *fact table* lists the facts posted and *speak up* asks members to share what others seem to lack; in the *lone dissenter* variant one member also holds a decisive fact. Blue: affordances; orange: prompt-only instructions. Mean ± bootstrap SE over the pairing units (10 seeds on the clue board, 100 episodes in the vote); * $p < 0.05$, two-sided sign-flip test of the paired difference against the baseline (exact over seeds; 10,000 random sign flips over episodes), uncorrected (the text gives the plan's Holm-corrected tests).

and in 1 a single update went wrong and was undone. The per-token estimate $e^d - d - 1$ (d: the reference's log-probability of a sampled token minus the policy's) grows exponentially when the policy makes a token far less likely than the reference does, so one token can dominate the logged mean and, through the loss, an update. With each token's term capped and two consecutive steps over the threshold required, none of the five later team-reward runs halted; we did not rerun the mixed-size runs with this fix.

4.5 Move information; do not add work

Channels and coordinators. A blackboard added 0.39 $[0.37, 0.40]$ welfare per agent-turn over no channel at four agents. A coordinator that collects clues and suggests subtasks added only 0.10. Against that coordinator, telling agents to follow its suggestions hurt (-0.13 $[-0.16, -0.10]$), as they selected subtasks for which they held no clue and waited for clues it never passed on, while having it pass on the clues it had collected helped (0.12 $[0.07, 0.16]$). A coordinator helps by moving information, not by giving orders.

Communication has a price. The baseline team earned a pooled share of 0.45 at four agents. Agents asked through the request board in 0.26 of their turns, yet sharing did not rise and the gap to the ceiling widened (gap closed -0.18 $[-0.21, -0.14]$ at four agents, Holm $p = 0.008$; about the same at 8, 16 and 32: -0.16, -0.17, -0.16). A clue table did not help (-0.01 $[-0.06, 0.04]$ at four agents, -0.05 at eight, Holm $p = 0.098$). An instruction to post every held clue first raised sharing from 0.40 to 0.73 shares per held clue but widened the gap (-0.14 $[-0.18, -0.09]$): each post costs a turn that could have gone to solving (Figure 5).

Public reputation, not private verification. With one liar among four agents, the track record recovered 0.32 $[0.20, 0.44]$ of the honest agents' loss (Holm $p = 0.008$), at no cost without a liar (0.006). The check action caught lies but recovered none of the loss (-0.01 $[-0.28, 0.19]$): a check cost a turn and informed only the checker. A warning did nothing (0.03). A 2×2 design crossing public or private with free or costly verification (10 seeds) found that making it public raised the share of loss recovered by 0.18 $[0.04, 0.36]$ and making it free by 0.16 $[0.02, 0.30]$: both matter, about equally. Unlike guards, the record needs the harness to know the truth.

5 Discussion

Where to put the work. The mechanisms that helped made information easier to move or to trust at no cost in turns (a shared board, a public record) or gave checking to agents whose job it was (guards); those that failed added work to every agent's turn (asking, checking, sharing first), demanded obedience, or only displayed what agents could infer for themselves (a clue table; agreement counts helped at 8 agents, no more than an instruction, and faded by 16). For harnesses this suggests: put verification in the environment or in a role, not in every agent's turn (on clean boards, asking every agent to confirm cost output and guards added nothing further); and split large teams into squads with their own guards rather than growing one global board, which limited the harm of the lies guards could not check.

From assigned to trained roles. Assigned roles worked; offered roles were not taken; and in five short runs team reward neither created roles nor kept every run productive. Roles will likely have to be trained, with a reward that pays an agent when others use its contribution, as credit reward did for sharing.

Future directions. Four experiments follow directly. (i) *Train roles rather than reward teams*: a role-conditioned or credit-style reward that pays a guard when its contest prevents a wrong answer, evaluated against the warm start. (ii) *Squads on coupled tasks*: subtasks whose clues span squads, with bridge agents that carry clues between them, to measure how much of the squads' gain survives when a task does not split cleanly. (iii) *Stronger adversaries*: Claude liars in 64-agent teams, liars that target the clues no guard holds, and free-text persuasion, against squads and failure flags. (iv) *Correlated errors and real tasks*: teams in which every copy shares a blind spot, guards drawn from a different model, and verification on coding or research tasks where a check costs more than a turn. We would start with (ii), the cheapest test of the squads result, and treat (i) as the main project. For all four, a simulator that captured what our rule-agent forecasts missed (Claude teams keeping their output as they grew, and acting on values the counting board had not confirmed) could rank designs before any API spend.

Limitations. The tasks are small and synthetic; the Claude teams use one model (Haiku 5.5) and mostly scripted faulty teammates; Claude liars could influence teammates only through clue posts; the plain board's rules told agents that re-sharing a clue only helps teammates who missed it, and every prompt listed the newest posted value among the clues an agent knew, which may have encour-

aged treating a posted clue as settled; the defences that ask agents to confirm also dropped that rule, a change not tested on its own; correlated errors shared by every copy of a model and compromised agents remain untested; and the saved runs and response cache are available on request, not in the public repository. Board arms have 5–10 seeds and RL designs one to eight training runs, which the failed replication (§4.4) shows is too few to rank similar designs; RL used a 3B model trained at four and eight agents only. The trust fixes have not been tested on a dedicated misinformation task such as the distributed fact recovery of Yan et al. [31].

Broader impact. The mechanisms studied are defensive or improve cooperation; the RL results show how easily a plausible incentive teaches agents to abandon cooperation, which matters for anyone training groups of models.

Author contributions

Chuntao Dan directed the project: chose the questions and the order of the studies, made the design decisions recorded in the specifications, ran the GPU training and reviewed every result. Claude Opus 5.5, working as an agent in Claude Code under that direction, wrote the specifications, analysis plans, environments and analysis code, ran the API experiments and replays, analysed the results and drafted the paper. Chuntao Dan takes responsibility for the content.

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A Reproducibility

Code, environments and notebooks: https://github.com/Positron00/explRL; a highlights page redraws the main figures from a compact extract of the saved runs (https://positron00.github.io/explRL/highlights.html). Analysis plans were frozen and committed before their runs (the coverage replay of §4.2 and the guessing counts of §4.4 were added after the results, with no frozen plan); each hash is the first 12 hexadecimal digits of the SHA-256 of the plan's canonical JSON (channels on the task board and in foraging: `ANALYSIS_PLAN`, hash `83df7591d1de`; the clue board: `CLUE_PLAN`, hash `050a825d7bae`; coordinator and instructions: `TRACKA_PLAN`, hash `45258d8a34da`; fixes: `FIX_PLAN`, hash `79abfb531cdd`; RL: `RL_PLAN` v2, hash `4d1ea0078a6e`; second RL round: `RL13_PLAN`, hash `3e50a4002c7b`; replication of the incentive result: `RL16_PLAN`, hash `082b3b7c9a31`; HiddenBench: `HB_PLAN`, hash `791a2df079a8`, frozen after a pilot, on the benchmark's 65 tasks vendored unchanged under their MIT licence; trust information: `TRUST_PLAN`, hash `6d2818ae84c2`; team size: `SCALE_PLAN`, hash `b886baefb2ef`, frozen after a pilot; traces: `TRACE_PLAN`, hash `f5c25637acd0`; guards: `COLONY_PLAN`, hash `58e842277a3e`; guards at 32 agents: `COLONY32_PLAN`, hash `f51f0298d34a`; guards at 64 agents: `COLONY64_PLAN`, hash `9764cd43e34c`; Claude liars and unassigned roles: `COLONY3_PLAN`, hash `0c59fa10f2b0`; guards that scale: `STAGE4_PLAN`, hash `a441381ce078`; the same designs at 32 agents: `STAGE4B_PLAN`, hash `4d5d0a8e4f5b`; team-size RL: `RL17_PLAN`, hash `8e28b5fdfb1f`; trained roles: `RL18_PLAN`, hash `c103584f5aac`). All API replies are cached locally (available on request), and cached reruns reproduce every row and turn log byte for byte at no cost. Every number in this paper is generated from the saved results (available on request with the cache) by `paper/make_assets.py`, and every figure by `tools/figures.py`.

B A vote fix that did not replicate

In a hidden-profile vote (four agents; shared facts favour the wrong option and each member holds two private facts favouring the right one; three rounds of posts, then a vote), members posted about half of their private facts (0.50) in every arm, yet a table of the facts posted so far closed 0.75 $[0.50, 1.00]$ of the gap when one member also held a decisive fact (win rate 0.88 to 0.97; Holm $p = 0.012$; Figure 5). Rerun unchanged on HiddenBench's 65 tasks [15] (195 episodes per arm), the table did not help (-0.07 $[-0.69, 0.30]$ of the gap); only an instruction to speak up, a control arm outside the Holm family, moved the outcome, and not significantly (gap closed 0.41 $[0.00, 0.71]$, $p = 0.09$): it raised disclosure of hidden facts from 0.74 to 0.88 and the win rate from 0.77 to 0.83. Our protocol left little headroom (agent accuracy 0.79 against the benchmark's reported 30%; one agent with every fact 0.91), and we do not claim the fact table's effect in general.

C A cheap simulation as a screen

Rule-agent simulations in numpy ranked harness designs the way Haiku teams did on a task board, where the risk is duplicate work (Spearman $\rho = 0.75$ $[0.55, 0.93]$), and on the clue board ($\rho = 0.79$ $[0.74, 0.81]$; 0.52 over the cells with a channel, since rule agents score zero without one), but not on foraging described in text ($\rho = 0.02$ $[-0.47, 0.45]$). Across the frozen forecasts, 14 of 21 effects were called right (sign matches for the reward forecast), but they missed the effects that depend on what language models do: guards paying off, larger teams losing more to liars, and guard copies winning at 64 agents (the design-change forecast was right on 4 effects, wrong on 1 and inconclusive on 1, where the Haiku interval spanned zero; at 32 agents it was right on 2, wrong on 0 and inconclusive on 1). A companion paper studies these simulations as a way to screen designs.